

34S Series — Metal Oxide Varistors

34 mm square, epoxy-encapsulated high-energy MOVs for surge protective devices · Datasheet · Rev. 2026-10-21

Description

The 34S series is the flagship line of 34 × 34 mm square, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.



Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top-grade 34S the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- High energy headroom: In 20 kA (25 kA for H variants), Imax up to 60 kA, 2 ms energy absorption up to 1,980 J.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Agency approvals

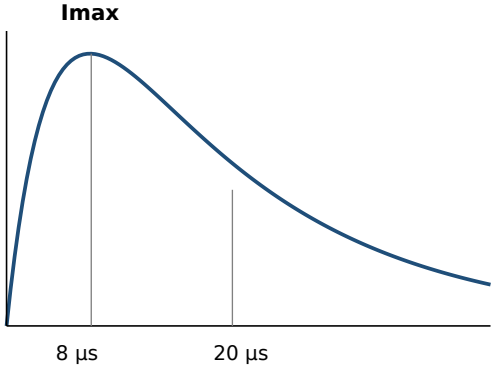
Standard	Category	File / status
UL 1449	VZCA2 (US) — surge protective components	File E335803, UL Recognized Component
CSA C22.2 No. 269.5-17	VZCA8 (Canada)	File E335803, UL Recognized Component

31 model grades are recognized, covering varistor voltages from 82 V to 1,800 V (model codes 820–182). Verify in UL Product iQ (productiq.ulprospector.com) by searching file number E335803 — the record lists exactly which model codes are covered. The UL Recognized Component Mark (mirrored “RU”) is etched on each recognized part together with the model code; a component carrying the circular UL Listing mark instead is a red flag worth verifying.

Absolute maximum ratings

Parameter	Range / limit	Unit
Continuous operating voltage, AC (VM(AC)rms)	60 to 1000	V
Continuous operating voltage, DC (VM(DC))	85 to 1465	V
Nominal discharge current In (8/20 µs)	20 (H variants 25)	kA
Maximum discharge current Imax (8/20 µs), standard range	up to 60	kA
Maximum energy absorption (2 ms)	up to 1,980 (single chip); up to 3,720 (multi-chip parallel, Section 4)	J
Operating temperature	−40 to +85 (no derating required)	°C

Impulse current waveform 8/20 µs (definition)



CAUTION: stresses above the ratings in the tables below may cause permanent damage. The ranges above summarize the per-model ratings in Sections 1–4; always design to the row for the exact model code.

Model naming and standard marking

Model naming: 34S K 431 — 34 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). /D = two-chip parallel, /T = three-chip parallel.

Standard marking (laser-etched on the encapsulated face, three lines): line 1 — the Flyboat lightning trademark with ®; line 2 — the model code, e.g. “34SK511” (34S series, ±10 % tolerance, 510 V varistor voltage); line 3 — the UL Recognized Component Mark followed by the production code, e.g. “26J209”.

1. Electrical characteristics — standard ratings

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK101	100	±10	60	85	20	40	360-3.6	190	✓	✓	✓
34SK121	120	±10	75	100	20	40	400-3.6	230	✓	✓	✓
34SK151	150	±10	95	125	20	40	450-3.0	290	✓	✓	✓
34SK181	180	±10	115	150	20	40	490-2.8	340	✓	✓	✓
34SK201	200	±10	130	170	20	40	520-2.6	390	✓	✓	✓
34SK221	220	±10	140	180	20	40	550-2.6	440	✓	✓	✓
34SK241	240	±10	150	200	20	40	600-2.6	460	✓	✓	✓
34SK271	270	±10	175	225	20	40	680-2.6	530	✓	✓	✓
34SK301	300	±10	195	250	20	40	770-2.6	600	✓	✓	✓
34SK331	330	±10	210	270	20	40	830-2.55	650	✓	✓	✓

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK361	360	±10	230	300	20	40	910-2.55	730	✓	✓	✓
34SK391	390	±10	250	320	20	40	1000-2.55	800	✓	✓	✓
34SK431	430	±10	275	350	20	50	1050-2.45	910	✓	✓	✓
34SK471	470	±10	300	385	20	50	1100-2.4	980	✓	✓	✓
34SK511	510	±10	320	410	20	50	1150-2.35	1050	✓	✓	✓
34SK561	560	±10	350	450	20	50	1250-2.35	1140	✓	✓	✓
34SK621	620	±10	385	505	20	50	1450-2.3	1240	✓	✓	✓
34SK621H	620	±10	385	505	25	60	1600-2.5	1280	—	—	—
34SK681	680	±10	420	560	20	50	1650-2.3	1250	✓	✓	✓
34SK681H	680	±10	420	560	25	60	1700-2.5	1280	—	—	—
34SK711	710	±10	440	585	20	40	1750-2.3	1260	✓	✓	✓
34SK751	750	±10	460	615	20	40	1800-2.3	1270	✓	✓	✓
34SK781	780	±10	485	640	20	40	1750-2.3	1280	✓	✓	✓
34SK821	820	±10	510	670	20	40	1800-2.3	1350	✓	✓	✓
34SK911	910	±10	550	745	20	40	2100-2.3	1460	✓	✓	✓
34SK951	950	±10	575	765	20	40	2200-2.3	1470	✓	✓	✓
34SK102	1000	±10	625	825	20	40	2400-2.3	1530	✓	✓	✓
34SK112	1100	±10	680	895	20	40	2700-2.3	1680	✓	✓	✓
34SK122	1200	±10	750	980	20	40	3000-2.3	1750	✓	✓	✓
34SK142	1400	±10	820	1120	20	40	3200-2.3	1780	✓	—	—
34SK152	1500	±10	850	1240	20	40	3400-2.3	1880	✓	—	—
34SK162	1600	±10	910	1320	20	40	3600-2.3	1920	✓	—	—
34SK182	1800	±10	1000	1465	20	40	4000-2.3	1980	✓	—	—

2. Electrical characteristics — low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
34SK431	430	±10	275	350	20	40	950-2.2	910	✓	✓	✓
34SK471	470	±10	300	385	20	40	1030-2.2	980	✓	✓	✓
34SK511	510	±10	320	410	20	40	1130-2.2	1050	✓	✓	✓
34SK561	560	±10	350	450	20	40	1240-2.2	1140	✓	✓	✓
34SK621	620	±10	385	505	20	40	1350-2.15	1240	✓	✓	✓
34SK681	680	±10	420	560	20	40	1480-2.15	1250	✓	✓	✓
34SK711	710	±10	440	585	20	40	1540-2.15	1260	✓	✓	✓
34SK751	750	±10	460	615	20	40	1630-2.15	1270	✓	✓	✓
34SK781	780	±10	485	640	20	40	1670-2.15	1280	✓	✓	✓
34SK821	820	±10	510	670	20	40	1850-2.1	1350	✓	✓	✓
34SK911	910	±10	550	745	20	40	1940-2.1	1460	✓	✓	✓
34SK951	950	±10	575	765	20	40	2040-2.1	1470	✓	✓	✓
34SK102	1000	±10	625	825	20	40	2200-2.1	1530	✓	✓	✓
34SK112	1100	±10	680	895	20	40	2380-2.1	1680	✓	✓	✓
34SK122	1200	±10	750	980	20	40	2620-2.1	1750	✓	✓	✓

3. Electrical characteristics — high-energy ratings (10/350 µs)

Model	U1mA (V)	UAC (V)	UDC (V)	In (kA)	Iimp (kA)	UP (V)	ETM (J)
34SK101	100	60	85	20	8	360-3.6	190
34SK121	120	75	100	20	8	400-3.6	230
34SK181	180	115	150	20	6	450-3.0	340
34SK201	200	130	170	20	6	490-2.8	390
34SK241	240	150	200	20	6	520-2.6	460
34SK271	270	175	225	20	6	550-2.6	530
34SK301	300	195	250	20	6	600-2.6	600

Model	U1mA (V)	UAC (V)	UDC (V)	In (kA)	Iimp (kA)	UP (V)	ETM (J)
34SK331	330	210	270	20	6	680-2.6	650
34SK361	360	230	300	20	6	770-2.6	730
34SK391	390	250	320	20	6	830-2.55	800
34SK431	430	275	350	20	6	910-2.55	910
34SK471	470	300	385	20	6	1000-2.55	980
34SK511	510	320	410	20	6	1050-2.45	1050
34SK561	560	350	450	20	6	1100-2.4	1140
34SK621	620	385	505	20	6	1150-2.35	1240
34SK681	680	420	560	20	6	1250-2.35	1250
34SK711	710	440	585	20	6	1450-2.3	1260

4. Electrical characteristics — multi-chip parallel ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
34SK431D	430	±10	275	350	30	65	1050-2.45	1460
34SK471D	470	±10	300	385	30	65	1150-2.45	1570
34SK511D	510	±10	320	410	30	65	1250-2.45	1680
34SK561D	560	±10	350	450	30	65	1350-2.4	1820
34SK621D	620	±10	385	505	30	65	1500-2.4	1980
34SK681D	680	±10	420	560	30	65	1650-2.4	2000
34SK711D	710	±10	440	585	30	65	1720-2.4	2010
34SK751D	750	±10	460	615	30	65	1820-2.4	2030
34SK781D	780	±10	485	640	30	65	1900-2.4	2050
34SK821D	820	±10	510	670	30	65	1990-2.4	2160
34SK911D	910	±10	550	745	30	65	2190-2.4	2340
34SK951D	950	±10	575	765	30	65	2210-2.4	2350
34SK102D	1000	±10	625	825	30	65	2420-2.4	2450
34SK112D	1100	±10	680	895	30	65	2660-2.4	2690
34SK122D	1200	±10	750	1060	30	65	2900-2.4	2800

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
34SK431T	430	±10	275	350	50	80	1130-2.6	2270
34SK471T	470	±10	300	385	50	80	1230-2.6	2450
34SK511T	510	±10	320	410	50	80	1340-2.6	2620
34SK561T	560	±10	350	450	50	80	1470-2.6	2850
34SK621T	620	±10	385	505	50	80	1630-2.6	3100
34SK681T	680	±10	420	560	50	80	1780-2.6	3120
34SK711T	710	±10	440	585	50	80	1860-2.6	3150
34SK751T	750	±10	460	615	50	80	1970-2.6	3170
34SK781T	780	±10	485	640	50	80	2040-2.6	3200
34SK821T	820	±10	510	670	50	80	2150-2.6	3370

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
34SK431D	430	±10	275	350	40	80	1120-2.6	1800
34SK471D	470	±10	300	385	40	80	1220-2.6	1960
34SK511D	510	±10	320	410	40	80	1320-2.58	2100
34SK561D	560	±10	350	450	40	80	1450-2.58	2280
34SK621D	620	±10	385	505	40	80	1600-2.58	2480
34SK681D	680	±10	420	560	40	80	1750-2.58	2500
34SK711D	710	±10	440	585	40	80	1830-2.58	2520
34SK751D	750	±10	460	615	40	80	1940-2.58	2540
34SK781D	780	±10	485	640	40	80	2010-2.58	2560
34SK821D	820	±10	510	670	40	80	2120-2.58	2700
34SK911D	910	±10	550	745	40	80	2350-2.58	2920
34SK951D	950	±10	575	765	40	80	2450-2.58	2940

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
34SK102D	1000	±10	625	825	40	80	2580-2.58	3100
34SK112D	1100	±10	680	895	40	80	2840-2.58	3300
34SK122D	1200	±10	750	1060	40	80	3100-2.58	3500

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)
34SK431T	430	±10	275	350	60	100	1150-2.7	2700
34SK621T	620	±10	385	505	60	100	1550-2.5	3720
34SK681T	680	±10	420	560	60	100	1700-2.5	3720

5. Dimensions and thickness

Bare electrode disc (–U): square chip, face 33 ± 0.5 mm. Bare-disc leaded forms: 34SKaaaDEN-U, 34SKaaaOBO-U, 34SKaaaZP-U, 34SKaaaFG-U (aaa = voltage-code digits). Encapsulated (epoxy-coated) forms: TH, CQ, WK, G1, WEL, KL, LD, LD2, LWB, M, YL, PR; multi-chip encapsulated: HL/D, LD/D, LD2/D, LWB/D, M/D, KL/D, WK/D, YL/D, TH/T, ZK/T. Single-chip encapsulated body ≈ 35–36 mm square (verify individual dimensions against the outline drawings before tooling).

Disc thickness Tmax (mm), bare disc, standard & low-clamping ranges — 101: 2.7 · 121: 2.7 · 151: 2.7 · 181: 2.8 · 201: 2.9 · 221: 3 · 241: 3.1 · 271: 3.3 · 301: 3.5 · 331: 3.6 · 361: 3.8 · 391: 4 · 431: 4.1 · 471: 4.4 · 511: 4.6 · 561: 4.9 · 621: 5.2 · 681: 5.2 · 711: 5.7 · 751: 6 · 781: 6.2 · 821: 6.4 · 911: 7 · 951: 7 · 102: 7 · 112: 7.6 · 122: 7.6

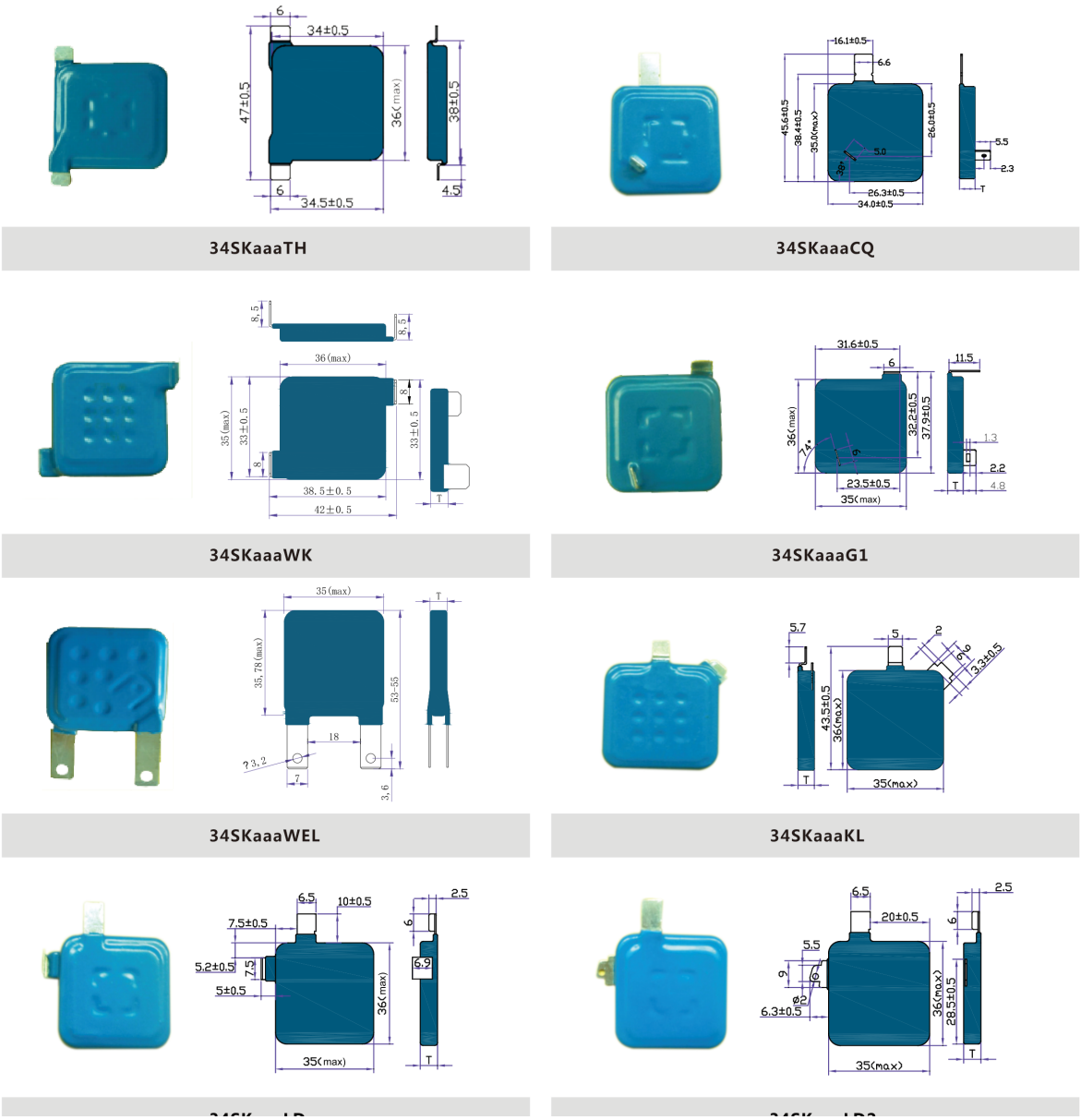
Disc thickness Tmax (mm), bare disc, high-energy (10/350) range — 101: 2.7 · 121: 2.7 · 151: 2.7 · 181: 3.3 · 201: 3.3 · 221: 4.2 · 241: 4.2 · 271: 4.9 · 301: 4.9 · 331: 5.5 · 361: 6.5 · 391: 6.5 · 431: 6.5 · 471: 8 · 511: 8 · 561: 9 · 621: 9 · 681: 9 · 711: 9

Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6 · 621: 6.3 · 681: 6.3 · 711: 6.6 · 751: 7.1 · 781: 7.3 · 821: 7.5 · 911: 8.1 · 951: 8.1 · 102: 8.1 · 112: 8.9 · 122: 8.9 · 142: 9.9 · 152: 10.4 · 162: 10.9 · 182: 11.9

Encapsulated thickness Tmax (mm), high-energy range — 431: 7.5 · 471: 7.5 · 511: 9 · 561: 9 · 621: 10 · 681: 10 · 711: 10

Multi-chip bare-disc thickness Tmax (mm) — /D two-chip — 431: 7.3 · 471: 7.9 · 511: 8.3 · 561: 8.9 · 621: 9.5 · 681: 9.5 · 711: 10.5 · 751: 11.1 · 781: 11.5 · 821: 11.9 · 911: 12.8 · 951: 13.1 · 102: 13.1 · 112: 14.3 · 122: 14.3

Multi-chip bare-disc thickness Tmax (mm) — /T three-chip — 431: 11.4 · 471: 12 · 511: 12.4 · 561: 13 · 621: 13.6 · 681: 13.6 · 711: 14.6 · 751: 15.1



34S encapsulated forms — outline drawings and dimensions (mm), from the company product catalog.

6. General specifications

- Operating temperature: −40 °C to +85 °C (no derating required); storage −40 °C to +125 °C (unloaded).
- Leakage current: ≤ 10 µA stable at 75 % of U1mA (DC).
- Ageing: after 1,000 h at 125 °C at 100 % DC load, ΔU1mA within −10 %, leakage ≤ 100 µA, no visible mechanical damage.
- Damp heat: 85 °C / 85 % RH, 2,000 h at UAC — ΔU1mA within −10 %, leakage ≤ 60 µA.
- RoHS: compliant with EU Directive 2011/65/EU. Solderability: lead-out terminals uniformly tinned.
- Marking / packaging: carton packing with packing list, outgoing inspection report and certificate of conformity; custom labels on request.

7. Encapsulated package forms

The 34S chip is supplied in a range of epoxy-encapsulated package forms with different terminal and mounting arrangements — eight standard forms are shown below. Outline drawings for each form are given in Section 5 above.



TH



CQ



WK



G1



WEL



KL



LD



LD2

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze. Certification marks in the ratings tables reproduce the catalog's own checkmark columns; verify coverage against the current UL Product iQ listing for file E335803 before external release.